Causal Inference Rework

The main variables of interest are `AlcoholRisk`, `SmokingDuration`, `NonSchoolQuals`, and `IncomeDecile`. I believe that all of them should remains as confounders (and some proxies, but this doesn't affect the model), based on the following sources and findings.

DAG Roles of Variables

Variable	Proposed DAG role	Rationale	Key sources
Smoking duration	**Potential confounder**	Smoking history is related to subsequent/current PA behaviour and independently relates strongly to GGT. Because the variable represents accumulated smoking history, it is temporally quite plausible for it to precede current PA status.	1, 3, 11
Alcohol risk level	**Potential confounder, but causal direction less certain**	Alcohol strongly predicts GGT/ALT and is associated with PA, but PA–alcohol associations vary between studies and direction is not completely established.	6, 7, 3
Non-school qualification	**SES proxy + upstream confounder**	Educational attainment is a component/proxy of SES; SES predicts PA and is also associated with GGT/liver-related health.	9, 10, 12
Income decile	**SES proxy + upstream confounder**	Income captures socioeconomic resources, which can influence opportunity for PA as well as health behaviours/metabolic/liver-health pathways.	9, 11, 12

Sources

1. Smoking and physical activity: a systematic review (source)
2. The effects of cigarette smoking and alcohol consumption on serum liver enzyme activities: a dose-related study in men (source)
3. Impacts of common factors of life style on serum liver enzymes (source)
4. Smoking and γ-glutamyltransferase: opposite interactions with alcohol consumption and body mass index (source)
5. Cigarette smoking and serum liver enzymes: the role of alcohol and inflammation (source)
6. Examining physical activity levels and alcohol consumption: are people who drink more active? (source)
7. Impact of Physical Activity on the Characteristics and Metabolic Consequences of Alcohol Consumption: A Cross-Sectional Population-Based Study (source)
8. Physical activity and the prevention, reduction, and treatment of alcohol and other drug use across the lifespan (The PHASE review): A systematic review (source)
9. Socio-economic determinants of physical activity across the life course: A "DEterminants of DIet and Physical ACtivity" (DEDIPAC) umbrella literature review (source)
10. Socioeconomic position and physical activity among women in Melbourne, Australia: does the use of different socioeconomic indicators matter? (source)
11. Longitudinal determinants of walking, moderate, and vigorous physical activity in Australian adults (source)
12. Socioeconomic Status, Antioxidant Micronutrients, and Correlates of Oxidative Damage: The Coronary Artery Risk Development in Young Adults (CARDIA) Study (source)